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Studierichting: Informatica
Oefeningenreeks 6C nr. 4.5

$$\begin{aligned}x_n &= \sqrt{n+1} - \sqrt{n} \\&= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\&= \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} \\&= \frac{1}{\sqrt{n+1} + \sqrt{n}} \\&\Rightarrow \lim_{n \rightarrow +\infty} x_n = \lim_{n \rightarrow +\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\infty + \infty} = \frac{1}{\infty} = 0 \\&\Rightarrow \text{adh}_{n \rightarrow \infty}(x_n) = \{0\} \\&\Rightarrow \liminf_{n \rightarrow \infty} x_n = 0 \\&\Rightarrow \limsup_{n \rightarrow \infty} x_n = 0\end{aligned}$$

References